



Applied Agentic AI – Foundation to Production (US Track)

For Walmart Global Tech · Global Tech Academy

Date: 16 June 2026 | Version: US – 20 Hours (2h Foundation + 18h Advanced)

Program at a Glance

This US track is designed for engineers who have some exposure to GenAI but need a rapid refresher before diving into production-grade decision-making.

To avoid redundancy and keep engagement high, we have crunched the foundational content into a single **2-hour accelerator**. The remaining **18 hours are entirely advanced**, focusing on architectural choices, evaluation, observability, cost optimization, and defence against review-board scrutiny – directly addressing Naman's feedback.

20h	5 Sessions	~70% Hands-on
Total duration (5 × 4 hrs)	2h Foundation + 18h Advanced	Lab over lecture

Audience

Built for **AI Engineers and Applied-AI Practitioners** who have built a few demos but now need to ship production systems that survive code reviews, cost governance, and incident post-mortems.

Dimension	Profile
Experience	4–10 years; has built prototypes (RAG, simple agents) but lacks production rigour
Current proficiency	Strong Python; comfortable with REST APIs; has used ChatGPT / Claude or internal LLMs
Environment	Mixed cloud (GCP / Azure); code-review gates; shipping-cadence pressure
Identity	Builders at scale – pragmatic, shipping-focused, ready to make defensible choices

Foundation Accelerator (2 Hours – Included in Session 1)

To ensure everyone starts from the same baseline, the programme begins with a 2-hour rapid refresher. This is **not** a deep dive; it is a targeted alignment exercise.

What is covered in these 2 hours:

- **LLM mechanics recap** (tokenization, context window, temperature) – brief, 20 mins
- **Prompt engineering patterns recap** (zero-shot, few-shot, chain-of-thought) – 30 mins
- **RAG architecture walkthrough** (chunking, embedding, retrieval) – 30 mins
- **Quick hands-on:** spin up a basic RAG pipeline using provided boilerplate – 40 mins

Note: This content is delivered at high velocity. Participants who are already comfortable with these basics will not feel bored, as they can skip straight to the lab exercise. Those needing a refresher get just enough to follow the advanced modules.

Prerequisite Knowledge (Strictly Assumed)

The following are **not** taught in this programme (beyond the 2-hour refresher). Participants must have working knowledge of these before joining, or complete pre-reads:

- Basic Python syntax and virtual environments
- How to call a REST API
- What a vector database is (conceptually)
- Basic understanding of LangChain agents (what a tool is, what a chain is)

Tools Required – Taught as Selection Judgement

Participants will use the following tools, but the focus is **not on tutorials** – it is on **when and why to choose one over another**.

Layer	Tool(s)
Language & runtime	Python (other languages referenced for comparison)
Orchestration frameworks	LangChain, LangGraph – taught comparatively (including when raw Python is better)
Agent interoperability	MCP (Model Context Protocol), function-calling / REST APIs – compared head-to-head
Embeddings & vector store	pgvector / Chroma / FAISS (lab); managed Pinecone for comparison
Evaluation & observability	Langfuse / Phoenix – golden-set regression, drift, cost / latency
Fine-tuning	LoRA / QLoRA via PEFT + Hugging Face; GPU-enabled Cloud Lab

Programme Structure – 5 Sessions × 4 Hours

The 20 hours are broken into 5 sessions of 4 hours each. Session 1 contains the 2-hour Foundation Accelerator + 2 hours of advanced content. Sessions 2–5 are fully advanced (16 hours).

Total advanced hours = 2 (Session 1) + 16 (Sessions 2–5) = **18 hours**.

Session 1 (4 Hours) – Foundation Accelerator + Engineering Decisions (Part 1)

Time Allocation	Content
0:00 – 2:00	Foundation Accelerator (rapid refresher, as described above)
2:00 – 4:00	Engineering Decisions – Part 1 (Advanced)

Outcome (Advanced Portion)

Participants begin justifying technology choices using trade-off matrices.

Advanced Topics Covered (2 hrs)

- **AI application architecture patterns** – Agent vs Workflow vs Traditional Software
- **RAG vs Fine-tuning vs Prompting** – quantified decision framework (cost, latency, quality, maintainability)
- **API vs MCP** – when to adopt MCP and when a REST API is sufficient

Lab (Hands-on)

- Analyse 2 business use cases and map them to the above technology choices.

Deliverable

Initial draft of a decision matrix for one use case (to be completed post-session).

Session 2 (4 Hours) – Engineering Decisions (Part 2) + Agent Architecture (Part 1)

Outcome

Participants can complete a full technology selection and design a basic agent system.

Engineering Decisions – Part 2 (2 hrs)

- **LangChain vs LangGraph vs Python-only** – complexity, maintainability, platform support, total cost of ownership
- **Build vs Buy** – harness systems, off-the-shelf agents vs in-house orchestration
- How to write a trade-off note that survives an architecture review

Agent Architecture – Part 1 (2 hrs)

- **Agent design principles** – single-agent vs multi-agent systems

- **Orchestration patterns intro:** Sequential, Router

Lab (Hands-on)

- Complete the Decision Matrix for the use cases from Session 1 (full deliverable).
- Implement a sequential agent and a router agent for a simple customer-support scenario.

Deliverable

Final Decision Matrix (structured document with clear trade-off rationale) + working code for two agent patterns.

Session 3 (4 Hours) – Agent Architecture (Part 2) + Evaluation & Observability (Part 1)

Outcome

Participants can design complex multi-agent orchestrations and understand how to measure their performance.

Agent Architecture – Part 2 (2 hrs)

- **Deep dive on orchestration patterns:** Planner-Executor, Supervisor / Hierarchical
- **Tool integration strategies** – built-in vs custom functions vs MCP servers
- **Agent-to-agent communication** protocols and failure scenarios (timeouts, hallucinations, context loss)

Evaluation & Observability – Part 1 (2 hrs) – as emphasised by Naman

- **Why evaluation matters** – moving from ad-hoc testing to rigorous measurement
- **Evaluation frameworks** – output-level (hallucination, toxicity), retrieval-level (hit rate, MRR), and agent-level (task success)
- Introduction to LLM-as-judge approaches

Lab (Hands-on)

- Take one use case and implement three different orchestration approaches (Planner, Supervisor, Router) using the same tools.
- Run a basic evaluation harness against them to compare accuracy and latency.

Deliverable

Architecture Comparison Report – side-by-side comparison of the three orchestration patterns.

Session 4 (4 Hours) – Evaluation, Observability & Debugging (Full Deep Dive)

Outcome

Participants can build production-grade evaluation pipelines and debug agentic workflows using observability data.

Topics (What & Why) – fully dedicated to Naman's key asks

- **Benchmark design** – creating golden datasets, regression testing, go / no-go gates
- **Logging architecture** – what to log, how to structure logs for agentic systems (traces, spans, tool calls)
- **Tracing and observability** – using Langfuse / Phoenix to visualise agent decision paths, identify bottlenecks, and detect drift
- **Debugging agentic workflows** – step-by-step root-cause analysis when a multi-step agent fails
- **LLM-as-judge in depth** – calibration, bias mitigation, and cost / accuracy trade-offs

Lab (Hands-on)

- Build a full evaluation pipeline for a RAG + agent system: run a golden test set, compute metrics, and generate a scorecard with a go / no-go gate.
- Instrument the same system with tracing and produce an observability dashboard that highlights latency, cost, and error rates per step.

Deliverable

Evaluation Scorecard + Observability Dashboard Design – a documented test harness with pass / fail thresholds, plus a working dashboard prototype.

Session 5 (4 Hours) – Production Readiness, AI Economics & Architecture Defence (Capstone)

Outcome

Participants can optimise operational costs, apply production-grade guardrails, and defend their full architecture in a simulated review board.

Topics (What & Why) – integrates Production Readiness and Economics / ARB

- **Production Readiness (1 hr):** guardrails, safety layers, prompt-injection defence, PII handling, scaling patterns, latency management, resilience (circuit breakers, retries, fallbacks).
- **AI Economics & Optimization (1 hr) – as emphasised by Naman:** token economics, context-window management, model routing, caching patterns, FinOps for GenAI, cost governance thresholds.
- **Architecture Defence & Capstone (2 hrs):** how to prepare for a formal review, what questions to expect, how to document trade-offs.

Capstone Lab (Integrated)

- Teams of 2–3 design a production AI solution for a given Walmart business problem.

They must present:

- **Architecture choice** – agent / workflow / traditional
- **Agent strategy** – orchestration pattern, tool selection
- **Evaluation strategy** – metrics, golden set, go / no-go
- **Cost model** – projected spend, optimisation levers
- **Deployment model** – CI/CD, rollback, incident response
- **Risk mitigation plan** – security, hallucinations, drift

Deliverable

Architecture Review Board (ARB) Presentation – a formal deck that defends every major architectural decision under cost, latency, and governance pressure. Peers and instructors act as the review board.

Final Programme Outcomes

By the end of the 20 hours, every participant will be able to:

#	Outcome
1	Select the right AI architecture for a business problem and justify it with a trade-off matrix.
2	Design agentic systems using appropriate orchestration patterns (sequential, router, planner, supervisor).
3	Build production-ready AI applications with security, reliability, and governance controls.
4	Evaluate, monitor, and debug AI systems using rigorous metrics, observability, and tracing.
5	Optimise token usage and operational costs via caching, model routing, and smart context management.
6	Defend technical decisions during formal architecture reviews, using data and documented trade-offs.

Assessment & Certification

Completion is based on active participation in labs and submission of **all deliverables**.

Certification requires a passing grade on the final ARB presentation (Session 5 capstone), assessed against a rubric covering:

- Architectural reasoning
- Trade-off documentation
- Cost / latency / governance awareness
- Risk identification and mitigation

No attendance-only certificates are issued.

Schedule Overview (Suggested)

Day	Session Content	Duration
Day 1	Session 1 – Foundation Accelerator (2h) + Engineering Decisions Part 1 (2h)	4 hours
Day 2	Session 2 – Engineering Decisions Part 2 (2h) + Agent Architecture Part 1 (2h)	4 hours
Day 3	Session 3 – Agent Architecture Part 2 (2h) + Evaluation & Observability Part 1 (2h)	4 hours
Day 4	Session 4 – Evaluation, Observability & Debugging (Full)	4 hours
Day 5	Session 5 – Production Readiness, AI Economics & ARB Capstone (Full)	4 hours